Recurrence exercises

1. Solve $a_0=3, a_n=2a_{n-1}-1$ for $n\geq 1$ by fixed-point substitution and iteration, checking both results.
2. Solve $a_n-5a_{n-1}+6a_{n-2}=0$ with $a_0=1, a_1=4$.
3. Solve $a_n-4a_{n-1}+4a_{n-2}=0$ with $a_0=2, a_1=6$.
4. Find the general solution of $a_n-2a_{n-1}=3^n$ and identify whether resonance occurs.
5. For $f(n)=2f(\lfloor n/2 \rfloor)+n, f(1)=1$, first restrict to $n=2^m$. Expand two levels, conjecture a closed form, and prove it by induction. Explain the effect of floors for general n.
6. Analyze $x_{n+1}=x_n+x_n^2$ for $x_0=0$ and $x_0=1$. Explain why characteristic roots do not apply and prove a lower bound showing unbounded growth in the second case.
7. Solve $b_0=1, b_n=4b_{n-1}+3n$ for $n\geq 1$. Find the homogeneous and polynomial particular solutions before fitting the initial value.
8. Solve $a_n=4a_{n-1}-4a_{n-2}+3\cdot 2^n$ for $n\geq 2$, $a_0=1, a_1=5$. Explain the forcing-base collision multiplicity and its required trial factor.
9. Solve $a_0=2, a_1=7, a_n=4a_{n-1}-3a_{n-2}$ for $n\geq 2$ by characteristic roots and OGF. Check the numerator, partial fractions, and initial values.
10. Use ordered binary trees to derive $C_0=1$ and $C_n=\sum_{i=0}^{n-1} C_i C_{n-1-i}$. Find C(x), derive $C_n=\binom{2n}{n}/(n+1)$, and check $C_4=14$. List the five stack output permutations for input 1,2,3 and explain why 312 is impossible.
11. Solve $a_{n+3}=2a_{n+2}$ for $n\geq 0$ with $a_0=3, a_1=4, a_2=5$. Identify zero-root multiplicity and a complete basis. Explain the failure of using only polynomial multiples of 0^n for zero-root terms.
12. With $T(1)=1$ and $T(n)=2T(n/2)+n/\log_2 n$ for $n=2^k\geq 2$, compute T(16) by layers, derive a general expression and asymptotic bound, and explain why polynomial-forcing master theorems cannot treat n/log n as n.
13. Solve $a_n=a_{n-1}-a_{n-2}$ for $n\geq 2$ with $a_0=1, a_1=0$. Give complex-root and real sine/cosine forms, compute seven terms, and prove period six.
14. For $a_0=1, b_0=0, a_n=2a_{n-1}+b_{n-1}, b_n=a_{n-1}+b_{n-1}$, build simultaneous OGF equations and derive A,B and both closed forms. Check through n=4. Prove the fixed-matrix formula $G=(I-xM)^{-1}y_0$ for $y_n=My_{n-1}$ even when M is singular.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study